BASIC PRINCIPLES OF ELECTRONICS

CHAPTER 3

T36NBV

"அணுவைத் துளைத்து ஏழ்கடலைப் புகட்டி
குறுகத் தரித்த குறள்". – ஔவையார்

Well before 2000 years, one of our women tamil poet and scholar "Avvaiyar" used the word 'அணு' i.e.atom, in her appreciation towards the universal religious book "Thirukkural". She used the word 'அணு' even before a single literal science or scientific word was derived. This proves the language Tamil and Tamilians were forerunners in deriving science, from basic to astronomy.

CONTENT

3.1 Atomic Structure
3.2 Structure of Elements
3.3 Electron Emission
3.4 Vacuum Tubes
3.5 Bohr's Atomic Model
3.6 Semi-conductor
3.7 Intrinsic Semiconductor
3.8 N-type Semiconductor
3.9 P-type Semiconductor
3.10 PN Junction

LEARNING OBJECTIVES

From this chapter the students would learn about the following:

1. Knowledge about basic electronic principles
2. Atomic structure of elements
3. Classification of Elements
4. Detailed knowledge of Semiconductors and its working
5. Working of PN-junction

INTRODUCTION

In this fast developing world, "Electronic" is the most important branch of engineering. Electronic devices are used in day to day common man life to big industrial activities. At its peak now robot are replacing human in areas where criticality and safety of human become risk.

The fast growth of this electronic technology offers a great challenge to the beginner, who likes to learn about electronics. This fundamental knowledge

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